

The role of scalar forcing in direct numerical simulations of turbulent atmospheric clouds

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Direct numerical simulation (DNS) is a promising approach to fill knowledge gaps regarding the microscale evolution of turbulent clouds. The governing equations for the flow of ambient air are solved in Eulerian framework, assuming homogeneous and isotropic turbulence. Two scalar fields, namely temperature and vapor mixing ratio, are transported by the turbulent velocity field. Aerosol particles and cloud droplets are tracked in Lagrangian fashion. It is known that forcing is required to sustain turbulence in numerical models. Although momentum forcing is common in DNS studies of clouds, forcing of the scalar fields has not gained enough attention. The present DNS model implements three types of scalar forcing: Fourier, mean gradient, and linear, in addition to momentum forcing. Our model's domain is initially divided into cloudy and clear-air regions, with mean scalar fields changing sharply at the interface, and subsequent mixing is simulated. The results suggest that forcing drives the mean scalar fields to equilibrium faster and suppresses microscale phase change. Scalar forcing counteracts the homogenization of scalar fields caused by flow turbulence and makes the probability distributions of forced scalar fields non-Gaussian. Scalar fields are coupled with particles in our model, and their spectra contain a viscous-convective regime which is further widened if forcing is applied. It is observed that mean gradient forcing produces extreme fluctuations in the scalar fields which translate into extreme microphysical responses, with a small fraction of cloud droplets experiencing super-adiabatic growth.

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NOMENCLATURE

α	= imposed scalar variance [K^2 for temperature and $(g/kg)^2$ for vapor mixing ratio]
β	= mean scalar gradient [K/m for temperature and g/kgm for vapor mixing ratio]
c_p, C_d	= specific heat of air [$J/(kg.K)$], rate of condensation or evaporation [$1/s$]
$\delta_{k,k_f}, \delta_T, \delta_{q_v}$	= delta functions [-] in the Fourier space, temperature field, and vapor field, respectively
$\epsilon_{T,in}, \epsilon_{q_v,in}$	= dissipation rates in the initial temperature [K^2/s], and vapor [$(g/kg)^2/s$] fields
ϵ, η	= turbulent kinetic energy dissipation rate [m^2/s^3], Kolmogorov scale [m]
η_B, E_T, E_{q_v}	= Bachelor scale [m], temperature spectrum [K^2m], vapor spectrum [$(g/kg)^2m$]
$\mathbf{F}_b, \mathbf{F}_c$	= buoyancy force [N], external force in the momentum field [N]
F_T, F_{q_v}	= external force in the temperature field [N], external force in the vapor field [N]
$\mathbf{g}, \mathbf{k}_f$	= acceleration vector due to gravity [m/s^2], forcing wavenumber vector [$1/m$]
k_{min}, k_{max}	= minimum wavenumber [$1/m$], maximum wavenumber [$1/m$]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [$W/(m.K)$]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m^2/s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m^2/s], instantaneous fluid pressure [N/m^2]
$q_v, q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
r_c, R_v	= critical radius [μm], specific gas constant for water vapor [$J/(kg.K)$]
$R_{q_v'q_v'}$	= auto-correlation coefficient for fluctuating water vapor mixing ratio [-]
ρ_a, ρ_l, S_e	= density of air [kg/m^3], density of water [kg/m^3], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
Sc, St, σ_l	= Schmidt number [-], Stokes number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
$\mathbf{u}, u_{rms}$	= instantaneous fluid velocity vector [m/s], root-mean-square velocity [m/s]
$\mathbf{V}, \mathbf{X}$	= particle velocity vector [m/s], particle position vector [m]
PDF, TKE	= probability density function [-], turbulent kinetic energy [m^2/s^3]

I. INTRODUCTION

Numerical modeling has become indispensable to study turbulence-cloud-aerosol interactions in the atmosphere since experimental measurements are few and difficult to conduct. The most accurate computational approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking. This method resolves the flow of turbulent air up to the smallest scale without any kind of turbulence modeling and tracks cloud droplets and aerosol particles individually in Lagrangian fashion.¹ Two thermodynamic (scalar) fields, namely temperature and vapor mixing ratio, are transported in Eulerian framework. DNS models rely on a forcing mechanism to maintain turbulent fluctuations.² Many DNS studies (Vaillancourt et al.,¹ Kumar et al.,³ and Yu Li et al.⁴) and our previous DNS model (Gao et al.⁵ and Sharfuddin et al.⁶) applied forcing in the momentum field only. While turbulence was sustained in the velocity field, fluctuations in the scalar fields decayed in those models. Thermodynamic fluctuations persist in natural clouds due to strong entrainment and mixing, convective motion, radiative transfer, and phase change⁷ while they decay otherwise. The objective of the present study is to simulate turbulence-cloud-aerosol interactions with sustained thermodynamic fluctuations. To achieve that, we apply external forcing in the scalar fields.

Without a turbulence production mechanism, turbulence decays in a numerical model and therefore an external forcing is needed. Eswaran and Pope² demonstrated that forcing in the Fourier space produces a statistically stationary velocity field. Janin et al.⁸ implemented linear forcing in the physical space to maintain homogeneous and isotropic turbulence (HIT). Fourier forcing is applied to a band of low wavenumbers in the Fourier space, but linear forcing is carried out in the physical space with energy injected at all nodal points. Besides momentum forcing, many previous studies have applied forcing in the scalar fields to prevent the decay of scalar fluctuations. Overholt and Pope⁹ applied mean gradient forcing in the physical space and showed that the fluctuating scalar field reaches a statistically stationary state. Gotoh and Watanabe¹⁰ implemented mean gradient forcing in a passive scalar field transported by HIT. Daniel et al.¹¹ proposed a forcing term in the physical space based on chemical reaction analogy and showed that this approach keeps the scalar field within a predefined bound. Chen and Cao,¹² Watanabe and Gotoh,¹³ and Kerr¹⁴ carried out scalar forcing in a band of low wavenumbers in the Fourier space. Carroll et al.¹⁵ adopted physical-space linear forcing that imposes a scalar variance and tested it for a range of Schmidt numbers. In the context of atmospheric clouds, Paoli and Shariff¹⁶ applied Fourier forcing in the temperature and vapor mixing ratio fields in addition to the momentum field. Chen et al.¹⁷ forced the scalar fields in a low wavenumber band to study the evolution of mixed-phase clouds.

Scalar forcing plays an important role at the small scales if scalar fields are coupled with particles. Satio et al.¹⁸ examined the relationship between scalar fluctuations and particles using a DNS model with Fourier scalar forcing. The authors reported that the TKE spectrum decays rapidly in the dissipation range but the scalar spectrum remains flat, suggesting the presence of a viscous-convective subrange. They attributed modification of the scalar spectrum to presence of particles that are smaller than the Kolmogorov length. Pumir¹⁹ used DNS and mean scalar gradient forcing for a range of Prandtl numbers to investigate the scale dependence of scalar fluctuations. The scalar spectra differ at all scales, including the small scales, and are different in the directions perpendicular and parallel to the mean gradient. However, the spectral profiles follow a slope of $-5/3$ and no flattening is observed in the inertial subrange due to absence of particles in their model. By applying a DNS model and Fourier scalar forcing, Watanabe and Gotoh¹³ studied the transport of a passive scalar field for which the Schmidt number is unity. The authors reported the existence of a viscous-convective subrange between the inertial and dissipation ranges. Although Schmidt number is less than unity for atmospheric flows, finite mass of cloud and aerosol particles can increase the effective Schmidt number.²⁰

Critical knowledge gap exists regarding the impact of scalar forcing in a DNS study of atmospheric clouds. So, we extend our previous DNS model⁶ to include forcing in the temperature and vapor mixing ratio fields. Three scalar forcing methods: Fourier, mean gradient and linear, are implemented. The fluctuating scalar fields are examined at the large, intermediate, and small scales. Previous DNS models^{9,10,13,15} with scalar forcing did not consider the presence of microscale particles. Also, our initial setup consists of cloudy and clear-air regions that are separated by a sharp interface unlike the models of Paoli and Shariff¹⁶ and Chen et al.¹⁷ We simulate condensation, aerosol activation into cloud droplets, evaporation, and cloud droplet deactivation into aerosols.⁶ The intensity of flow turbulence is varied. The relationship between velocity turbulence, scalar forcing, and cloud microphysics are discussed.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations, the scalar transport equations, the equations for scalar forcing in the physical and Fourier spaces, and the equations for Lagrangian particle tracking.

A. Turbulent velocity field

The turbulent air in the atmosphere is assumed to be governed by the physical-space Navier-Stokes equations with the Boussinesq approximation:²¹

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Here, $\mathbf{u}$ is the instantaneous fluid velocity vector and p is the instantaneous fluid pressure, $\mathbf{F}_b$ is the buoyancy force, and $\mathbf{F}_c$ is the external force applied to sustain flow turbulence. The definitions of $\mathbf{F}_b$ and $\mathbf{F}_c$ can be found in Ref. 6.

B. Scalar transport and scalar forcing

The thermodynamic variables or scalar quantities in this problem are temperature, water vapor mixing ratio, and supersaturation. The transport equations for temperature and water vapor mixing ratio fields are written in the Eulerian framework as:⁵

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T + \delta_T F_T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v + \delta_{q_v} F_{q_v}. \quad (4)$$

where C_d quantifies the time rate of mass exchange between liquid water and water vapor and is described by Eq. (24). C_d is positive during condensation and negative during evaporation. The terms $L_h C_d / c_p$ and $-C_d$ are latent heat terms in Eqs. (3) and (4), respectively. F_T and F_{q_v} are the forcing terms for temperature and water vapor mixing ratio, respectively; and δ_T and δ_{q_v} are the delta functions defined as

$$\delta_T = \begin{cases} 1 & \text{if } \frac{L_h}{c_p} C_d > 0 \\ -1 & \text{if } \frac{L_h}{c_p} C_d < 0 \end{cases}. \quad (5)$$

$$\delta_{q_v} = \begin{cases} 1 & \text{if } -C_d > 0 \\ -1 & \text{if } -C_d < 0 \end{cases}. \quad (6)$$

δ_T and δ_{q_v} ensures that the forcing terms act in the direction of corresponding latent heat terms.

We implement three scalar forcing methods. The first is Fourier forcing for which the forcing term is constructed in the Fourier space as:²²

$$\hat{f}_\theta(\mathbf{k}, t) = \epsilon_\theta \frac{\hat{\theta}(\mathbf{k}, t)}{\sum_{k_f \in k} |\hat{\theta}(\mathbf{k}_f, t)|^2} \delta_{k, k_f}. \quad (7)$$

$$\epsilon_\theta = \epsilon_{\theta, in} m_\theta \quad (8)$$

Above, θ denotes the scalar field, $\hat{\theta}(\mathbf{k}, t)$ is the Fourier-transformed scalar field, $\epsilon_{\theta, in}$ is the dissipation rate in the initial scalar field and m_θ is the multiplication factor. Forcing is applied in the wavenumber band of $1 \leq k_f / k_{min} \leq$

4.12, where k_f is the forcing wavenumber and k_{min} is the minimum wavenumber. The second method is mean gradient forcing for which the forcing term is defined in the physical space as:⁹

$$F_{\theta, MG} = w\beta \quad (9)$$

where w is the instantaneous z-velocity and β is the mean scalar gradient. The unit of β is for K/m for temperature forcing and g/kgm for vapor forcing. The β is calculated by matching $F_{\theta, MG}$ with $\hat{f}_{\theta}^{-1}(\mathbf{k}, t)$ as:

$$\beta = \left\langle \frac{\hat{f}_{\theta}^{-1}(\mathbf{k}, t)}{w} \right\rangle \quad (10)$$

Note that $\hat{f}_{\theta}^{-1}(\mathbf{k}, t)$ is found by transforming $\hat{f}_{\theta}(\mathbf{k}, t)$ back to the physical space. The third method is linear scalar forcing and the forcing term is constructed in the physical space as:¹⁵

$$F_{\theta, LN} = \frac{1}{2} \frac{\langle \chi(\Theta) \rangle}{\sigma_{\Theta}^2} \Theta \quad (11)$$

The rescaled scalar field Θ is defined as¹⁵

$$\Theta = \theta \sqrt{\frac{\alpha}{\sigma_{\theta}^2}} \quad (12)$$

where α is the imposed scalar variance. The quantity $\chi(\Theta)$ is the dissipation rate of Θ and is defined as¹⁵

$$\chi(\Theta) = \frac{\partial \sigma_{\Theta}^2}{\partial t} \quad (13)$$

We find α as follows:

$$\begin{aligned} \hat{f}_{\theta}^{-1}(\mathbf{k}, t) &= \frac{1}{2} \frac{\langle \chi(\Theta) \rangle}{\sigma_{\Theta}^2} \Theta = \frac{1}{2} \frac{\langle \chi(\Theta) \rangle}{\sigma_{\Theta}^2} \theta \sqrt{\frac{\alpha}{\sigma_{\theta}^2}} \\ \alpha &= \left(\frac{2 \hat{f}_{\theta}^{-1}(\mathbf{k}, t) \sigma_{\theta}^2}{\langle \chi(\Theta) \rangle \theta} \right)^2 \sigma_{\theta}^2 \end{aligned} \quad (14)$$

Note that α is updated at every grid point and every timestep, whereas β is updated at every timestep but is constant in space. In our model, the β and α could not be specified a priori as shown in Refs. 9 and 15 because we have two scalar fields with different mean values. Hence, we match mean gradient and linear forcing functions with Fourier forcing function.

We solve the combined (mean plus fluctuating) scalar fields as shown in Eqs. (3) and (4). Since forcing affects both the fluctuating and mean scalar fields, we define scalar forcing functions in a way that avoids unphysical results. For example, a multiplication factor is used in Eq. (8) so that Fourier forcing is neither too weak nor too strong. It is

expected that the mean supersaturation decreases and increases to zero (saturation level) during condensation and evaporation, respectively. Our experience shows that a strong forcing can cause the mean supersaturation to be less or greater than zero, which would be unphysical. On the other hand, a weak forcing cannot sustain scalar fluctuations at a desired level. The multiplication factor for temperature (m_T) is larger than that for vapor mixing ratio (m_{q_v}) (Table III) because $\langle T \rangle > \langle q_v \rangle$, where $\langle . \rangle$ indicates spatial averaging. Also, the scalar forcing term (F_θ) interacts with the latent heat term. If the magnitude of former is larger than that of latter with opposite sign, condensation might take place instead of evaporation and the vice versa, altering the physics of the problem. To prevent that, the delta functions have been introduced in Eqs. (5) and (6). Note that δ_T and δ_{q_v} will augment the temporal changes in the T and q_v by aligning the forcing functions with the direction of phase change terms. This supports the physical scenario in which the T and q_v evolve rapidly due to intense mixing between cloud and environment.²³

C. Growth of cloud droplets and aerosol particles

The motion of each aerosol particle or cloud droplet is represented in the Lagrangian fashion. The simplified expressions for the Lagrangian position and velocity are⁵

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad (15)$$

$$\frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}. \quad (16)$$

where the subscript ' i ' denotes the i -th aerosol particle or cloud droplet, $\mathbf{V}$ is its velocity vector, and $\mathbf{X}$ is its position vector. $\mathbf{u}_i$ is the instantaneous fluid velocity vector at particle position i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by⁵

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \quad (17)$$

In the atmosphere, inter-conversion of aerosol particles and cloud droplets takes place.²⁴ When the environment is supersaturated, aerosol particles grow beyond the critical radii and activate into cloud droplets. Whereas cloud droplets shrink below the critical radii and deactivate into aerosol particles in a subsaturated environment. The critical radius (r_c) and critical equilibrium supersaturation (S_c) can be derived from the kappa-Kohler theory as:⁶

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}, \quad S_c = \frac{2}{\sqrt{\kappa}} \left(\frac{A}{3r_d} \right)^{\frac{3}{2}}. \quad (18)$$

where r_d is the dry aerosol radius, r is the particle radius, κ is the hygroscopicity of the solute, and A is a parameter defined in Ref. 6. A particle can be classified as aerosol particle or cloud droplet by comparing r with r_c . An aerosol particle corresponds to $r_d < r \leq r_c$ while a cloud droplet corresponds to $r > r_c$. With r_d known from the given dry aerosol population, r_c can be calculated using Eq. (18). The change in particle (aerosol or cloud droplet) radius during condensation (or evaporation) is described by²⁵

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)} (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (19)$$

where S_e is the environmental supersaturation and S_k is the particle equilibrium supersaturation that accounts for the curvature and solute effects combined.⁶ The S_e is defined as

$$S_e = \frac{q_v}{q_{v,s}} - 1, \quad (20)$$

where $q_{v,s}$ is the saturation water vapor mixing ratio.²⁶ The S_e is an Eulerian quantity in Eq. (20) but interpolated to the particle position in Eq. (19). The growth parameter G is expressed as²⁵

$$G = \frac{1}{\frac{L_h \rho_l}{k'_T T} \left(\frac{L_h}{R_v T} - 1 \right) (1 + S_k) + \frac{\rho_l R_v T}{\mu'_v p_{sat}}}, \quad (21)$$

where k'_T and μ'_v are modified thermal conductivity and modified water vapor diffusivity in air,²⁷ respectively. With the foregoing, the C_d is determined as:

$$C_d(\mathbf{x}, t) = \frac{4\pi\rho_l}{\rho_a a^3} \sum_{i=1}^n G_i r_i(t) (S_e(\mathbf{X}, t) - S_k(\mathbf{X}, t))_i. \quad (22)$$

where n is the number of particles in a grid cell that has a volume of a^3 .

III. NUMERICAL DETAILS

We conduct two studies: Study I and Study II. As explained in Ref. 6, the first study is based on the scenario that condensation and activation dominate in one physical region of a cloud while the second study resembles the scenario that evaporation and deactivation prevail in another physical region. The domain length (0.512 m) of our model is much smaller compared to an actual cloud since DNS is computationally expensive. As a consequence, no natural mechanism (convection, Dirichlet boundaries, or radiative effects) is available in our model to prevent the decay of turbulence. Therefore, external forcing is required in the momentum and scalar fields to sustain turbulent fluctuations.

A. The numerical model

By varying the range of forced wavenumbers,⁶ we generate two turbulence levels: ‘low’ and ‘high’, and denote them as ‘L’ and ‘H’, respectively. Specifically, we apply forcing in the wavenumber bands: $1 \leq k_f/k_{min} \leq 4.12$ and $1 \leq k_f/k_{min} \leq 12.12$, respectively. Note that the domain length $L \equiv 0.512 \text{ m}$, the minimum wavenumber $k_{min} = 2\pi/L \sim 12.27$ and the maximum wavenumber $k_{max} \sim 128k_{min}$ in our model. The initial TKE spectrum is identical for both turbulence levels and is specified as²⁸

$$E(k) = \frac{16}{\sqrt{\pi/2}} \frac{u_0^2 k^4}{k_0^5} \exp\left(-\frac{2k^2}{k_0^2}\right). \quad (23)$$

where $k_0 = 4.12k_{min}$ and $u_0 = 0.16 \text{ m/s}$ is the root-mean-square (rms) velocity. Table I lists the turbulence statistics such as the turbulent kinetic energy, $\bar{h}$, its dissipation rate, ϵ , the rms velocity, u_{rms} , the Kolmogorov length scale η , the Taylor micro-scale, λ , the length scale characterizing large eddies, l_0 , the forcing Reynolds number, Re_f , the Taylor-scale Reynolds number, Re_λ , and the Reynolds number based on l_0 , Re_{l_0} . The abovementioned quantities are defined in Ref. 6. The integral length scales⁶ (l) are 0.14 m and 0.08 m for ‘low’ and ‘high’ turbulence levels, respectively. The parameters listed in Table I are representative of turbulence in real clouds.²⁹

Table I. Statistics of the turbulent velocity field at 4 s

Turbulence level	$\bar{h}$ (m ² /s ²)	ϵ (m ² /s ³)	u_{rms} (m/s)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.0094	0.00494	0.1386	0.0169	0.00091	0.184	401.12	156.11	1704.7
Low	0.0025	0.00055	0.0797	0.0261	0.00157	0.227	192.97	138.74	1207.6

The physical model simulated is that of a cubic domain (box) inside which turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, and the domain volume is 0.512^3 m^3 . At the start of simulations, the domain is divided equally into cloudy and clear-air regions. Cloud droplets or aerosol particles are initially placed in the cloudy region and subsequent turbulent mixing causes them to spread to all over the domain. A slab-like cloud configuration⁶ is assumed for the initial field (Fig. 1), to enable a sharp transition³⁰ in the water vapor mixing ratio and temperature at the cloud/clear-air interface. Such an inhomogeneous distribution of the initial scalar fields allows us to realistically simulate turbulent mixing between cloud and environment.

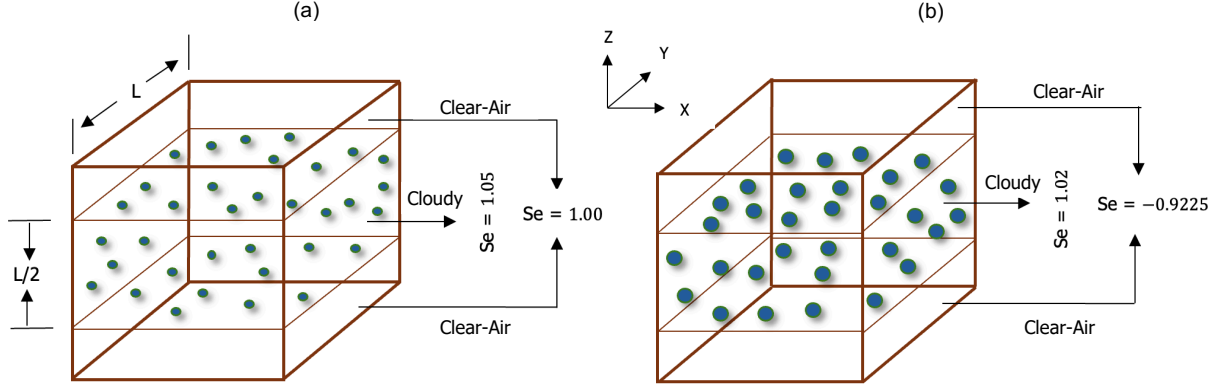


FIG. 1 Initial configurations for (a) Study I, and (b) Study II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. The cloudy region is initially 5% supersaturated in Study I and 2% in Study II. The clear-air region is just saturated in the first and -92.25% subsaturated in the second. The spheres represent aerosol particles in Study I and cloud droplets in Study II.

In Study I, the cloudy region is supersaturated (humid), but the clear-air region is just saturated at initial time. The initial vapor mixing ratio field is specified as:⁶

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases} \quad (24)$$

The value of $q_{v,s}$ in Eq. (20) is 3.872 g/kg . Above, $q_v^{max} = 1.05 \times q_{v,s} = 4.066 \text{ g/kg}$, and $q_{v,e} = 3.872 \text{ g/kg}$. The length, $d \equiv L/2$ is the width of the cloud slab. The temperature field is initialized as:⁶

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (25)$$

To resolve the discontinuous distribution (Eq. (24)), we choose a DNS model based on finite difference⁶ instead of a pseudospectral DNS model because the latter could give rise to artificial oscillations (Gibbs phenomenon).³¹ At the start of simulations, 16×10^6 dry aerosols are randomly placed in the cloudy region, giving a number concentration of 119 cm^{-3} . We simulate six cases: L, L-F, L-MGF, L-LF, H, and H-F, which are summarized in Table II. The identifiers ‘L’ and ‘H’ before the dash (-) imply ‘low’ and ‘high’ levels of flow turbulence, respectively. The identifier ‘F’ after the dash (-) indicate that scalar forcing is applied in the Fourier space while ‘MGF’ and ‘LF’ denote mean gradient forcing and linear forcing, respectively. Note that momentum forcing is implemented in the Fourier space for all cases. The size distributions of dry aerosol particles are assumed to be lognormal with geometric mean radius of $0.126 \mu\text{m}$ and standard deviation of $0.02 \mu\text{m}$. Dry aerosols (solutes) absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Study I.

TABLE II. Cases in Study I and Study II. Case L has ‘low’ turbulence level but without scalar forcing. Case L-F has ‘low’ turbulence level and scalar forcing is implemented in the Fourier space. Case L-MGF applies mean scalar gradient forcing in the physical space and calculates the β adaptively. Case L-LF implements linear scalar forcing in physical space and calculates the α adaptively. Cases H and H-F are analogous to Cases L and L-F except that the turbulence level is ‘high’.

Case	Dry aerosol size (μm)	TKE dissipation rate (m^2/s^3)	Turbulence level	Scalar forcing type	Initial particle size in Study I (μm)	Initial particle size in Study II (μm)
L	0.126 ± 0.02	0.0025	Low	N/A	0.126 ± 0.02	10
L-F	0.126 ± 0.02	0.0025	Low	Fourier	0.126 ± 0.02	10
L-MGF	0.126 ± 0.02	0.0025	Low	Mean gradient	0.126 ± 0.02	10
L-LF	0.126 ± 0.02	0.0025	Low	Linear	0.126 ± 0.02	10
H	0.126 ± 0.02	0.0094	High	N/A	0.126 ± 0.02	10
H-F	0.126 ± 0.02	0.0094	High	Fourier	0.126 ± 0.02	10

In Study II, the cloudy region is supersaturated (humid) but the clear-air region is subsaturated (dry) initially. The same configurations for vapor mixing ratio and temperature as in Eqs. (24) and (25) are used except that $q_v^{max} = 1.02 \times q_{v,s} = 3.949 \text{ g/kg}$ and $q_{v,e} = 0.30 \text{ g/kg}$. A total of 16×10^6 cloud droplets, having a monodisperse size (radius) distribution of $10 \mu m$, are randomly placed in the cloudy region at the start. Also, dry aerosols (r_d) are present in the core of cloud droplets (r). Note that r_d and r_c at each particle position do not change with time but r evolves with time. We simulate the six Cases (Table II) in Study II. Cloud droplets evaporate, shrink in size, and deactivate into aerosol particles in Study II.

Particles have zero velocity at the initial time. The particle growth equation (Eq. (19)) is solved after 2 s. By that time, the TKE dissipation rate has become statistically stationary. The dissipation rates in the initial temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{q_v,in}$) fields are listed in Table III. The Schmidt number ($Sc = \nu/\mu_T$) is 0.68 for both temperature and vapor mixing ratio. The Bachelor length ($\eta_B = \eta/\sqrt{Sc}$) are 0.0019 m and 0.0011 m for ‘low’ and ‘high’ turbulence intensities, respectively. However, the effective Sc is higher in our model because scalar fields are coupled with particles via latent heat terms. To balance the grid resolution requirement and computational cost, we carry out all the simulations using 256^3 grid points. The $k_{max}\eta$ values for ‘low’ and ‘high’ turbulence levels are 2.46 and 1.43, respectively.

The FronTier software package³² is used to perform calculations. The details about discretization and numerical techniques can be found in Ref. 6. We leverage the Perlmutter supercomputer managed by National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA to run our simulations.

B. List of parameters

We list the constants and parameters of our DNS model in Table III. The hygroscopicity parameter (κ) is 0.61, assuming ammonium sulfate composition for dry aerosols.³³

TABLE III. Initial parameter values and constants

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Study I)	g/kg	$q_{v,e}$	3.872 (Study I)	g/kg
	3.949 (Study II)			0.30 (Study II)	
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	1.0	kg/m ³	ν	1.5×10^{-5}	m ² /s
ρ_l	1000	kg/m ³	p	82844.14	N/m ²
$\langle T(t=0) \rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m
$\mu_T = \mu_v$	2.16×10^{-5}	m ² /s	M_l	0.018	kg/mol
R	8.314	J/mol/K	S_t	7.35×10^{-8} (Study I)	-
κ	0.61	-		4.63×10^{-4} (Study II)	
$\epsilon_{T,in}$	2.16×10^{-5} (Study I)	K ² /s	$\epsilon_{qv,in}$	7.9×10^{-4} (Study I)	(g/kg) ² /s
	7.7×10^{-3} (Study II)			2.8×10^{-1} (Study II)	
m_T	10 (Study I)	-	m_{qv}	0.1 (Study I)	-
	1 (Study II)			0.1 (Study II)	

C. Model validation

We run our model using the parameters of Overholt and Pope⁹ who applied mean scalar gradient forcing. The time evolution of normalized scalar variance is plotted in Fig. 2(a). The Cases OP-1996 and OP-128 in Fig. 2(a) represent Overholt and Pope⁹ (Run 128.3) and our run, respectively. The number of grid points is 128^3 for both cases.

The scalar variance σ_θ^2 is defined as:

$$\sigma_\theta^2 = \frac{\sum_i (\theta_i - \bar{\theta})^2}{N}. \quad (26)$$

where N is the number of grid points. The variance is zero initially for both cases and no latent heat term is present in the scalar transport equation. The scalar variance is normalized by $(\beta L_\epsilon)^2$, where L_ϵ is a turbulent length scale.⁹ The fluctuating scalar field is normalized by its standard deviation σ_θ , and the normalized time is t/T_E , where t is the physical simulation time and T_E is the eddy turnover time.⁹ We find that the normalized scalar variances for Cases

OP-1996 and OP-128 (Fig. 2(a)) are almost equal once they become statistically stationary ($t/T_E \sim 6$). The differences observed between 2 to 6 (abscissa) is attributed to the transient behavior in respective models.

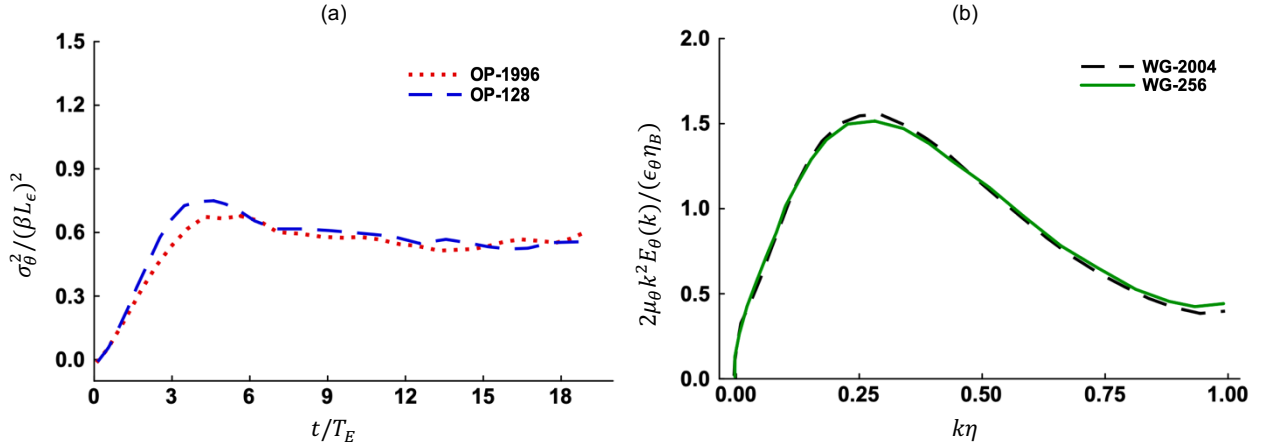


FIG. 2 Comparison of (a) normalized scalar variances and (b) normalized scalar dissipation spectra.

We also validate our model against that of Watanabe and Gotoh¹³ who carried out scalar forcing in the Fourier space. According to Ref. 13, the normalized scalar dissipation spectrum is: $2\mu_\theta k^2 E_\theta(k) / (\epsilon_\theta \eta_B)$, where μ_θ , E_θ , and ϵ_θ are diffusivity, spectrum, and dissipation rate in the scalar field θ , respectively. Figure 2(b) shows the normalized scalar dissipation spectra for Cases WG-2004 and WG-256 which represent Watanabe and Gotoh¹³ (Run 1) and our run, respectively. The number of grid points is 512^3 for the former but 256^3 for the latter. We find excellent agreement between the two profiles in Fig. 2(b).

IV. RESULTS AND DISCUSSIONS

The results from our simulations are presented in this section. We examine the effects of scalar forcing on the fluctuating and mean scalar fields during condensation and evaporation as well as on the activation and deactivation processes. The impact of different flow turbulence intensities on forced and unforced scalar fields and microphysics is also discussed, followed by the concluding remarks.

A. The impact of forcing on the fluctuating and mean scalar fields

1. Condensational growth (Study I)

Study I is concerned with condensational growth and aerosol activation. The temporal evolution of the variances of temperature (σ_T^2), vapor mixing ratio (σ_{qv}^2), and environmental supersaturation (σ_{se}^2) is shown in Figs. 3(a)-(c), respectively. We see that the forced scalar fields attain statistical stationarity, but the unforced scalar fields (Case L) continue to decay. The mean gradient forcing term, unlike the linear forcing term, is not matched with the Fourier

forcing term at every nodal point. Rather, the mean gradient β is found by spatial averaging (Eq. (10)). Hence, the magnitudes of fluctuations are largest for Case L-MGF but very close for Cases L-F and L-LF. The origin of scalar fluctuations in our model is the sharp scalar gradients that exist at the initial cloud/clear-air interface. With time, the fluctuating scalar fields decay due to mixing if scalar forcing is absent. Whereas scalar forcing acts as a source of fluctuations and counteracts the decay.

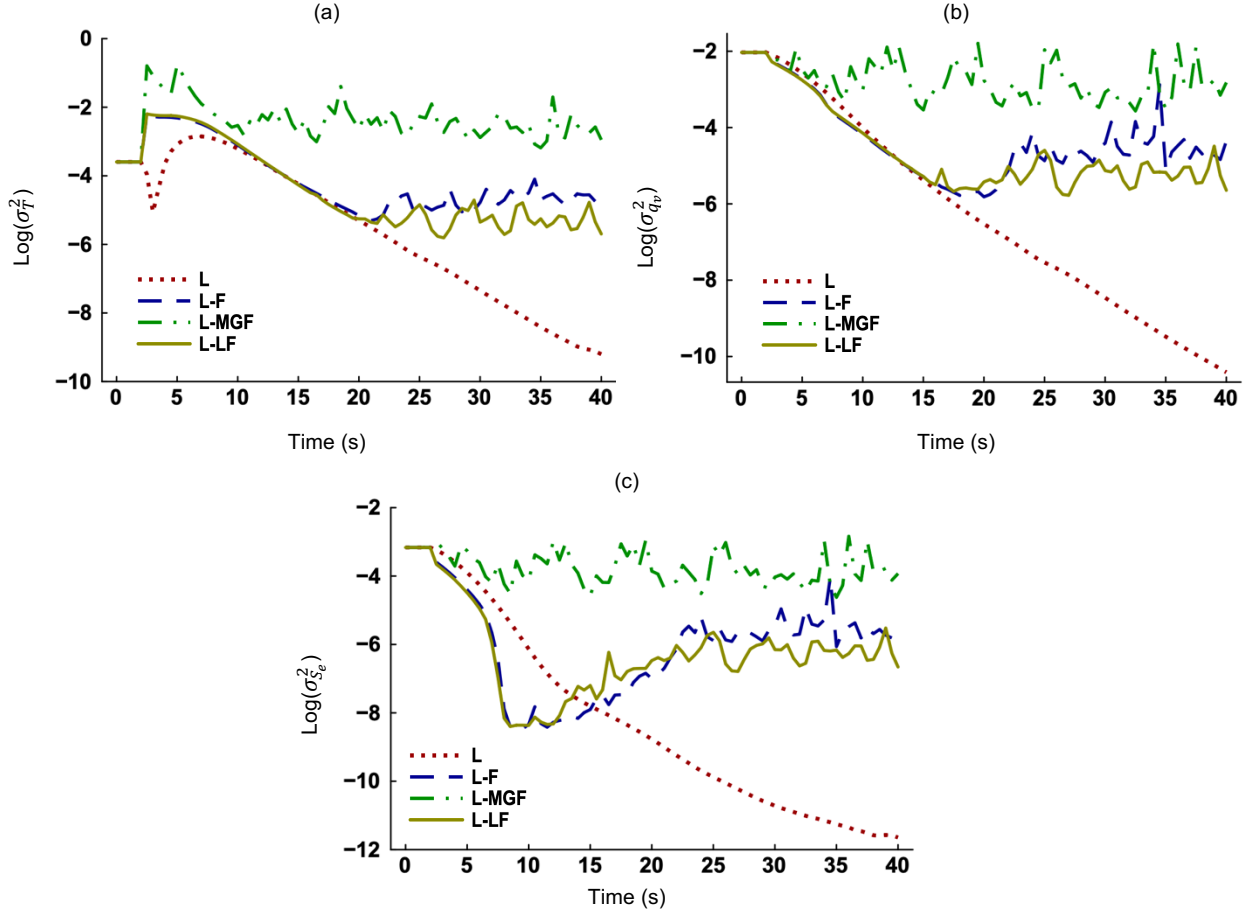


FIG. 3 Time evolution of the variances of (a) temperature, (b) vapor mixing ratio, and (c) environmental supersaturation, in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and calculates the β adaptively. Case L-LF has linear scalar forcing and calculates the α adaptively. The turbulence level is ‘low’ for all cases.

The spectra³⁴ of temperature and vapor mixing ratio are plotted in Figs. 4(a)-(b). They show the distribution of scalar fluctuations across scales. According to the Kolmogorov’s theory, small-scale structures should be universal and independent of the large-scale anisotropy.³⁵ However, the spectra do not collapse at the small scales in Figs. 4(a)-(b). They differ at the intermediate and small scales between Cases L and L-F while those between Cases L and L-MGF differ at all scales. A flat region ($3.0 \leq \log(k) \leq 3.2$) can be identified in the spectra for Case L, which suggests the existence of a viscous-convective regime, typically observed for scalars with Schmidt number (Sc) greater than

unity.³⁶ The Sc for both scalars are less than unity (0.68) in our model if particles are excluded. But the coupling of scalar fields with tiny cloud droplets and aerosol particles increases the Sc because transport of mass (particles) is slower than transport of momentum. The flat region expands further ($1.8 \leq \log(k) \leq 3.2$) for Cases L-F and L-LF, and the viscous convective regime appears to dominate the inertial convective regime for these two forced cases. Consequently, the spectral slopes deviate from the $-5/3$ law in the inertial range as seen in Figs. 4(a)-(b). A key feature of high Sc is the increase of high-wavenumber contribution to the scalar spectrum.³⁷ We find that fluctuations increase at the intermediate and small scales or high-wavenumber regions for Cases L-F and L-LF compared to the unforced case (L). When mean gradient forcing is applied (Case L-MGF), a constant (mean) scalar variance is injected at all physical-space nodal points and fluctuations increase at all scales compared to Case L.

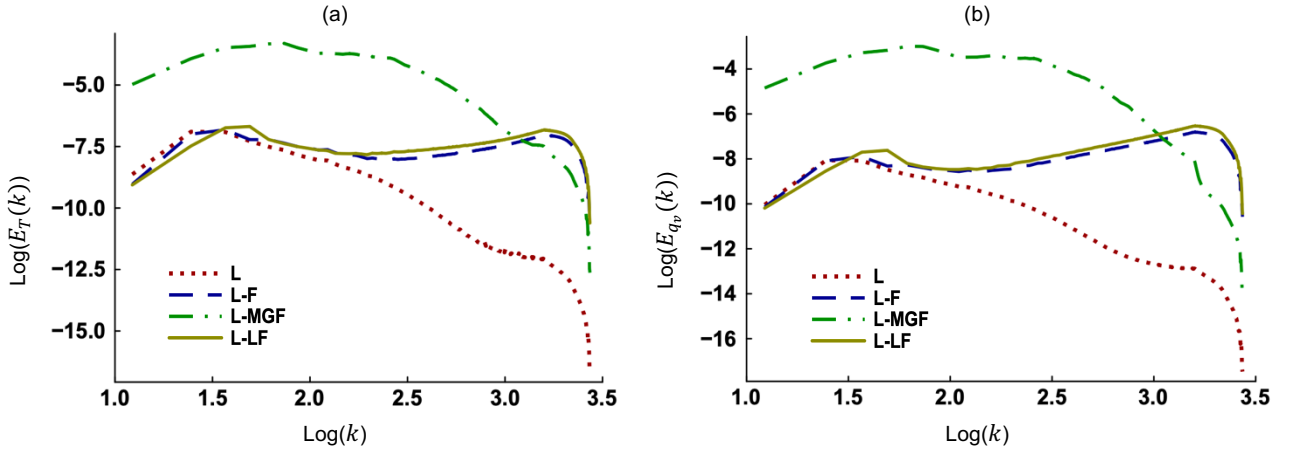
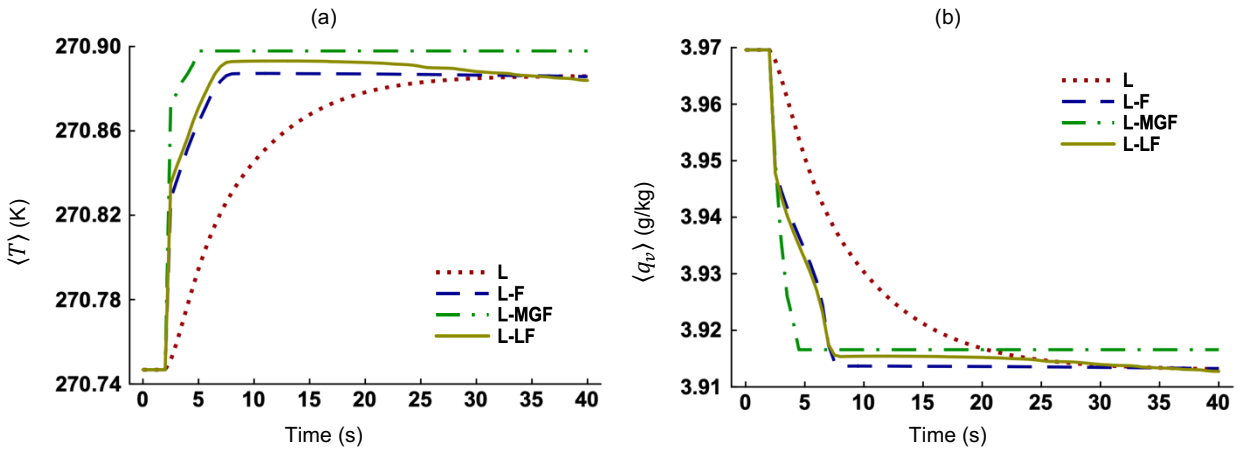


FIG. 4 Spectra of the (a) temperature and (b) vapor mixing ratio fields in a log-log format. The results represent Study I at 25 s.



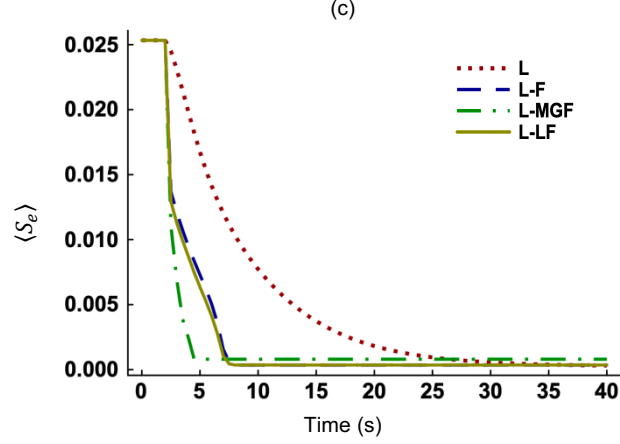


FIG. 5 Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is ‘low’ for all cases.

Figures 5(a)-(c) show the temporal evolution of mean scalar fields in Study I. The temporal increase in the $\langle T \rangle$ and decrease in the $\langle q_v \rangle$ and $\langle S_e \rangle$ are slowest when scalar forcing is absent (Case L). Conversion of water vapor into liquid water during condensation reduces the $\langle q_v \rangle$ and thus the $\langle S_e \rangle$. Figure 5(c) suggests that the system reaches equilibrium when the $\langle S_e \rangle$ approaches zero (saturation). Equilibrium is reached by approximately 35 s for Case L and 7 s for the forced cases. We find that forcing causes the mean scalar fields to relax to their equilibrium states faster. This is because phase change drives the evolution of mean scalar fields in our model (Fig. 6(a)) and scalar forcing acts in the direction of phase change.

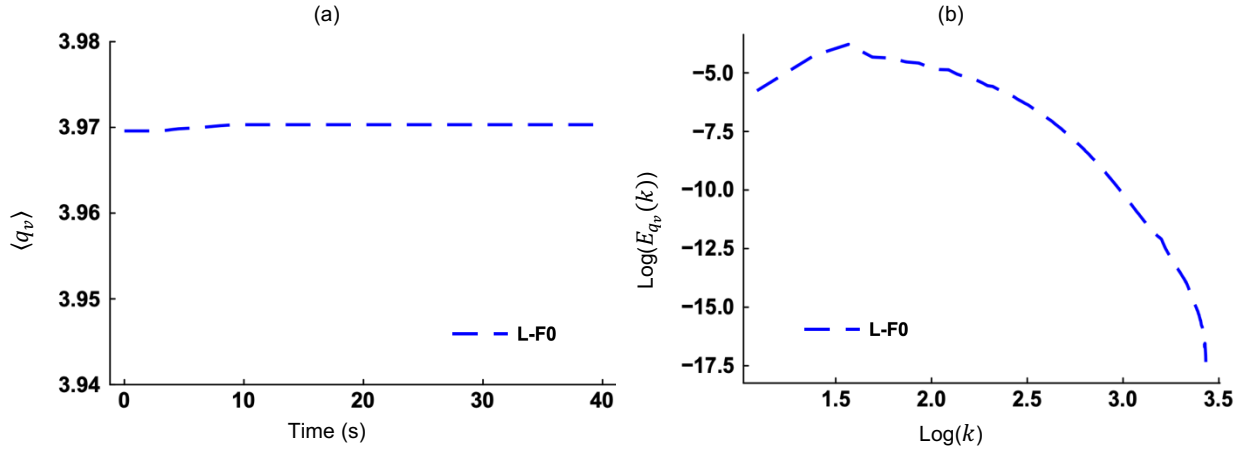


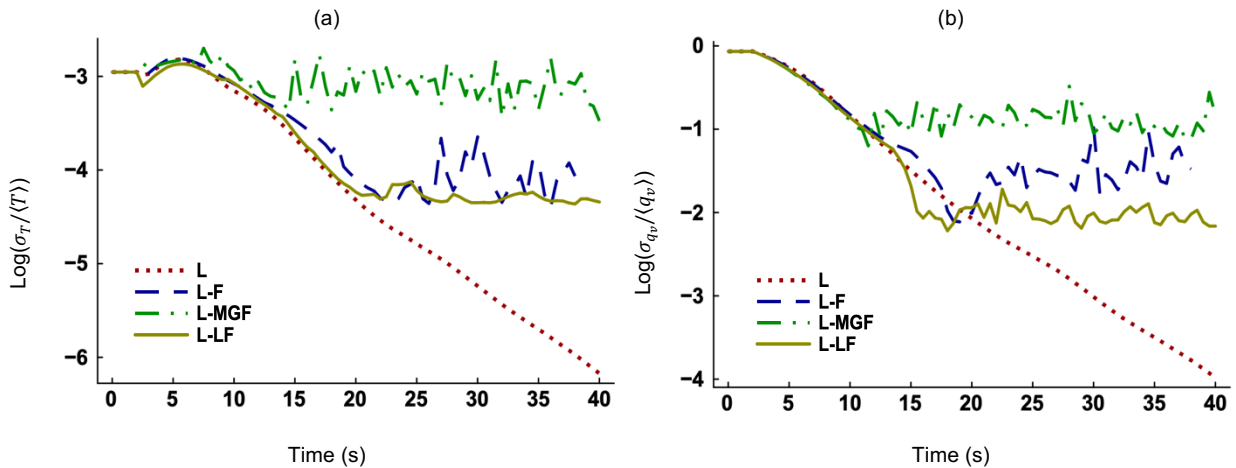
FIG. 6 (a) Time evolution of the mean vapor mixing ratio ($\langle q_v \rangle$) and (b) spectrum of the vapor mixing ratio ($E_{q_v}(k)$) at 10 s in log-log format. The results are for Case L-F0 in Study I. Case L-F0 implements Fourier scalar forcing but excludes particles.

To recognize the connection between scalar fields and particles, we simulate Case L-F0 by setting the particle size (r) zero. When r is zero, C_d is also zero (Eq. (22)) and so are the latent heat terms in Eqs. (3) and (4). We see in

Fig. 6(a) that the temporal variation in the $\langle q_v \rangle$ is negligible for Case L-F0. Since the exclusion of particles prevents phase change, water vapor cannot turn into liquid water during condensation. So, the $\langle q_v \rangle$ remains unchanged with time. Figure 6(b) shows the spectrum of vapor mixing ratio (E_{q_v}) at 10 s for Case L-F0. There is a peak in the $E_{q_v}(k)$ profile, followed by an exponential decay at the small scales as expected in HIT.² This suggests that Fourier forcing itself does not cause deviation from homogeneity and isotropy in the scalar fields when particles are absent. But the interaction between scalar forcing and inertial particles modifies the scalar spectra as shown for Cases L-F, L-MGF, and L-LF (Figs. 4(a)-(b)).

2. Evaporative decay (Study II)

Study II focuses on evaporative decay and cloud droplet deactivation. The temporal evolution of the relative dispersions of scalar fields ($\sigma_\theta/\langle\theta\rangle$) is plotted in Figs. 7(a)-(c). The fluctuating scalar fields decay in absence of forcing but become statistically stationary if forcing is applied. The $\sigma_T/\langle T \rangle$ and $\sigma_{q_v}/\langle q_v \rangle$ for Case L at 25 s in Study II are $1.6e-5$ and $2.97e-3$, respectively. The corresponding values for Case L-F are $8.5e-5$ and $2.73e-2$. Hence, fluctuations in temperature and vapor mixing ratio increase with forcing. Their standard deviations are much smaller than the means, which is also the case for natural clouds.³⁸ So, forced temperature and vapor mixing ratio fields remain within practical bounds. The $\sigma_{S_e}/\langle S_e \rangle$ is $8.78e-3$ for Case L and $9.04e-3$ for Case L-F in Study II at 25 s. The σ_{S_e} is larger than the $\langle S_e \rangle$ with Fourier forcing but the opposite is true for the unforced case (L). This finding is consistent with the result of Paoli et al.¹⁶ who applied scalar forcing in the Fourier space. Because the $\langle S_e \rangle$ is much smaller than the $\langle T \rangle$ and $\langle q_v \rangle$, fluctuations in supersaturation are more pronounced with forcing.



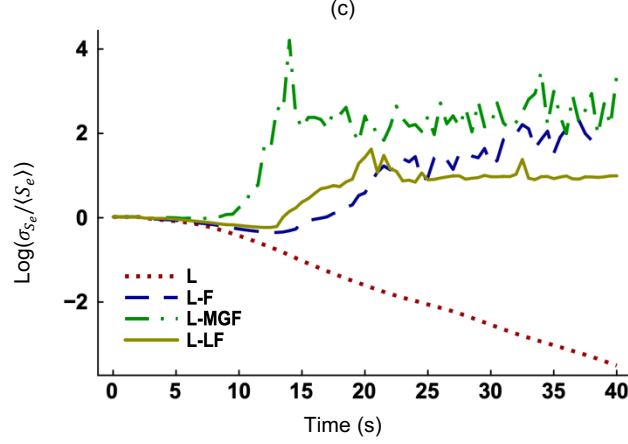


FIG. 7 Time evolution of the relative dispersions of (a) temperature, (b) vapor mixing ratio, and (c) environmental supersaturation, in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean gradient forcing and Case L-LF has linear forcing. The turbulence level is ‘low’ for all cases.

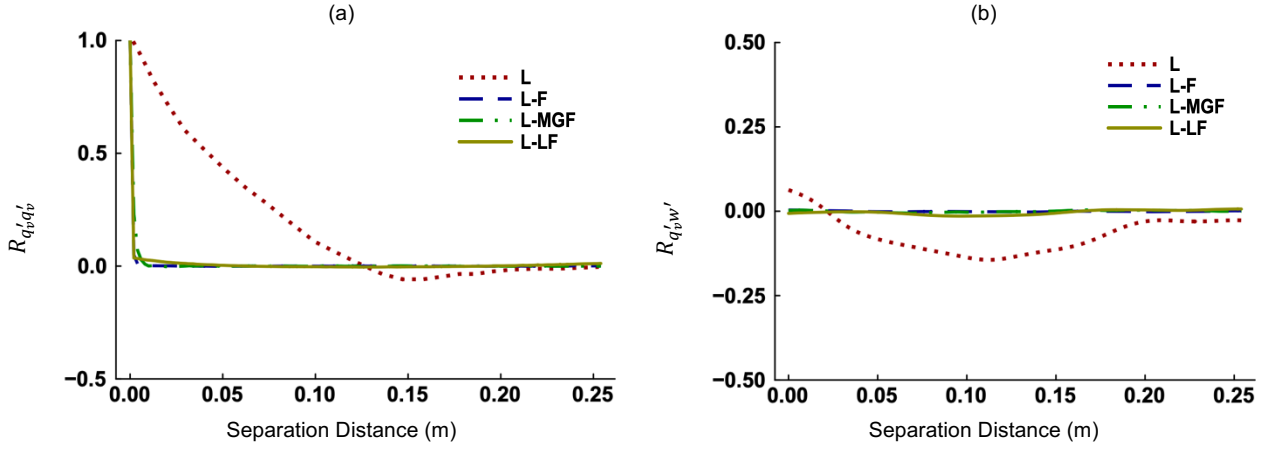


FIG. 8 The correlation coefficients (a) $R_{q'_v q'_v}$ and (b) $R_{q'_v w'}$ in Study II at 25 s.

Spatial correlation coefficient quantifies the mutual dependence between two quantities separated in space. The autocorrelation coefficient $R_{q'_v q'_v}$ is defined as³⁴

$$R_{q'_v q'_v}(\mathbf{r}) = \frac{\langle q'_v(\mathbf{x}, t) q'_v(\mathbf{x} + \mathbf{r}, t) \rangle}{\sqrt{\langle (q'_v(\mathbf{x}, t))^2 \rangle \langle (q'_v(\mathbf{x} + \mathbf{r}, t))^2 \rangle}}, \quad \mathbf{r} = \Delta x \mathbf{i} + \Delta y \mathbf{j} + \Delta z \mathbf{k}. \quad (29)$$

where $\mathbf{r}$ is the separation distance vector and q'_v is the fluctuating vapor mixing ratio. The cross-correlation coefficient $R_{q'_v w'}$ can be defined similarly with w' being the fluctuating z-velocity. The $R_{q'_v q'_v}$ and $R_{q'_v w'}$ are plotted in Figs. 8(a)-(b), respectively. We find that $R_{q'_v q'_v}$ decays almost instantaneously for Cases L-F, L-MGF, and L-LF compared to the unforced case (L). The reason is that fluctuations in the scalar fields are driven by locally injected scalar variances. So, the values of q'_v are not correlated with each other if separation distance increases. The $R_{q'_v q'_v}$ gradually approaches

zero for Case L when a source of scalar fluctuation is absent. We see in Fig. 8(b) that $R_{q'_v w'}$ is essentially zero and thus q'_v and w' are not correlated at all for the forced cases. This confirms that fluctuations in the forced scalar fields are not driven by flow turbulence. In absence of scalar forcing (Case L), q'_v and w' are weakly correlated because the velocity field merely transports the scalar fields in our problem and does not augment or dissipate the latter.⁶

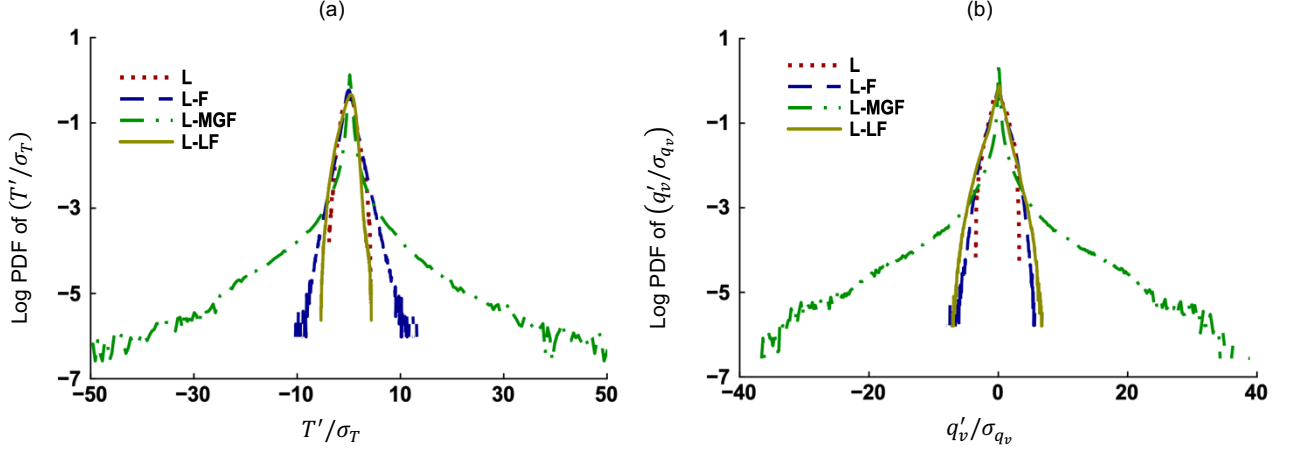


FIG. 9 Log PDFs of the (a) fluctuating temperature (T') normalized by standard deviation (σ_T), and (b) fluctuating vapor mixing ratio (q'_v) normalized by standard deviation (σ_{q_v}), at 25 s in Study II.

Figure 9(a) shows the PDFs of T'/σ_T , which is the fluctuating temperature normalized by standard deviation, at 25 s for the four cases. Figure 9(b) is analogous to Fig. 9(a) but for vapor mixing ratio. It appears that the normalized PDFs in both figures are broader for Case L-MGF compared to the other cases. To assess how much Gaussian the distributions are, we calculate their skewness (γ) and kurtosis (ω) as³⁵

$$\gamma = \frac{\langle(\varphi')^3\rangle}{\langle(\varphi')^2\rangle^{3/2}}, \quad \omega = \frac{\langle(\varphi')^4\rangle}{\langle(\varphi')^2\rangle^{4/2}}, \quad \varphi = \frac{\theta'}{\sigma_\theta}. \quad (30)$$

The γ of temperature PDFs in Fig. 9(a) are 0.055, 0.28, -0.12, and -0.76 for Cases L, L-F, L-MGF, and L-LF, respectively, while the corresponding ω are 3.21, 6.68, 425.2, and 3.73. The γ of vapor PDFs in Fig. 9(b) are 0.09, 0.003, -0.035, and -0.31, respectively, while the corresponding ω are 3.03, 4.43, 176.2, and 5.84. Note that skewness is 0 and kurtosis is 3 for a perfectly Gaussian distribution. The kurtosis is close to 3 for Case L but is larger than 3 for the forced cases. Thus, forcing makes the PDFs of fluctuating scalar fields non-Gaussian. Previous studies of cloud turbulence found that thermodynamic fluctuations can be both Gaussian and non-Gaussian.^{7,39} With mean gradient forcing (Case L-MGF), the PDFs in Figs. 9(a)-(b) have heavy tails and their kurtosis are very high.⁴⁰ Hence, extreme values are observed for this case although most of the values are closer to the mean. During updrafts and downdrafts, extreme fluctuations in temperature, vapor mixing ratio, and supersaturation have been reported.⁴¹

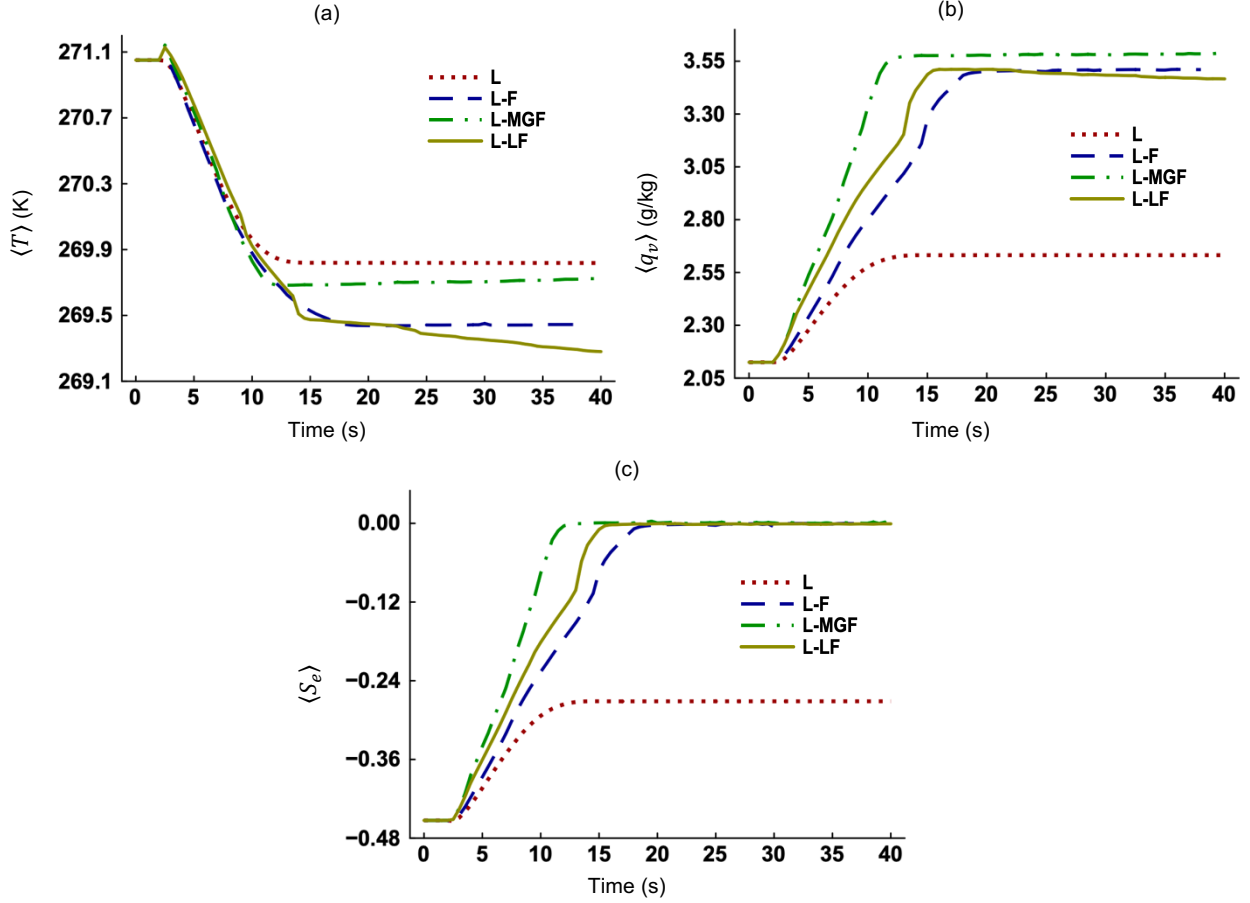


FIG. 10 Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is ‘low’ for all cases.

The temporal evolution of the mean scalar fields for Study II is shown in Figs. 10(a)-(c). We see that the growth and decay profiles are opposite to those in Figs. 6(a)-(c). During evaporation, liquid water converts into water vapor, increasing the $\langle q_v \rangle$ and $\langle S_e \rangle$. The decrease in $\langle T \rangle$ is a manifestation of evaporative cooling. Equilibrium is reached before saturation ($\langle S_e \rangle \sim 0$) for Case L (Fig. 10(c)). Absence of scalar forcing (Case L) leads to faster evaporation and the profile of $\langle S_e \rangle$ settles to an asymptote when all droplets evaporate fully. On the other hand, equilibrium corresponds to saturation for the forced cases and is reached before evaporation is complete.

B. The impact of scalar forcing on microphysics

1. Activation of aerosol particles (Study I)

The number fraction of aerosol particles that activate into cloud droplets with time is plotted in Fig. 11(a). The activation process is very fast between 2 s to 6 s for Case L, slows down afterwards, and eventually all aerosol particles

activate. For the forced cases, the activation fraction is lower because equilibrium is reached before aerosol particles could grow sufficiently to be activated. Interestingly, the activation profile for Case L-MGF decreases slightly after its peak at around 4 s. In this case, a small number of particles deactivate after activation. Those particles experience negative growth due to negative fluctuations in the S_e (Eq. (19)) and deactivate when r falls below r_c . Figure 11(b) show the time evolution of the mean radius. The profiles tend to asymptote after exponential increase at early times. This aligns with the prediction of Eq. (19) that particle growth rate (dr/dt) slows down as r increases and S_e decreases (Fig. 5(c)). For the unforced case (L), particle growth (Fig. 11(b)) and activation fraction (Fig. 11(a)) are the highest. Scalar forcing drives the system towards equilibrium faster (Fig. 5), which leaves less time available for completion of phase change.⁴² Hence, particle growth (condensation) is reduced, and activation is suppressed.

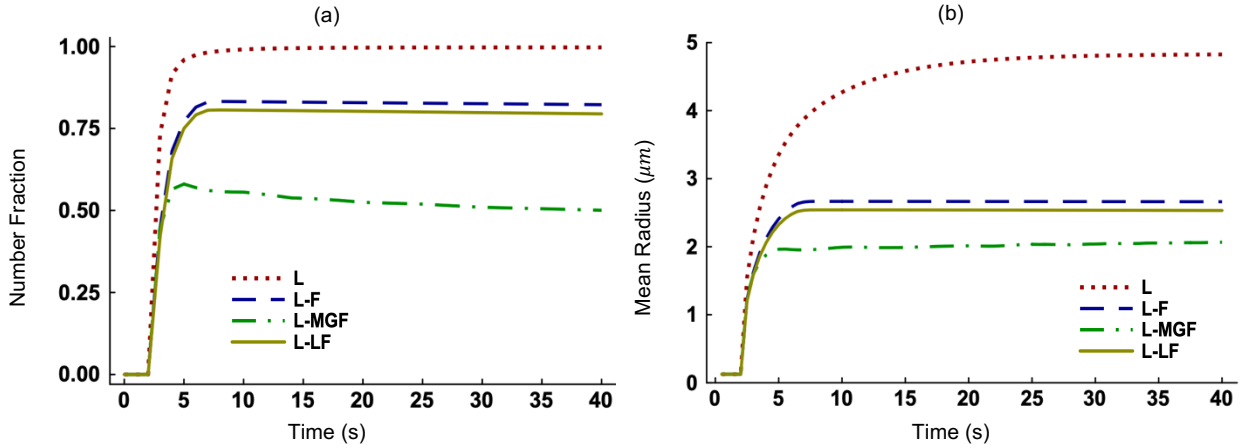
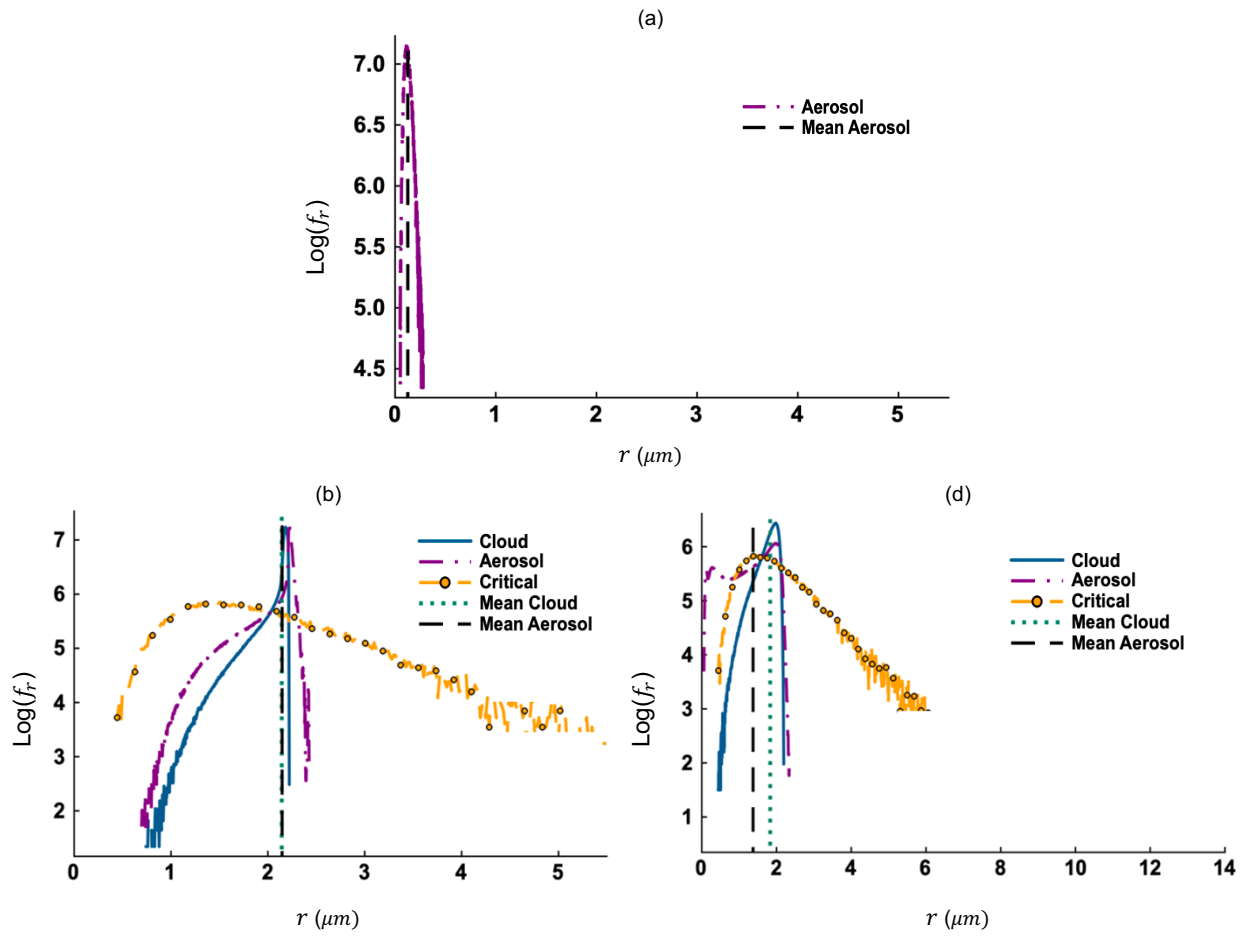


FIG. 11 Time evolution of the (a) number fraction of activated aerosol particles, and (b) mean radius of all particles, in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is ‘low’ for all cases.

Cloud and aerosol size distributions differ for the unforced and forced cases. Figures 12(a)-(c) show the PDFs of particle radii (f_r) for Case L during condensation (Study I) at 0 s, 3 s, and 10 s, respectively. Figures 12(d)-(e) are analogous to Figs. 12(b)-(c) but represent Case L-MGF. Lognormally distributed dry aerosol particles are present at 0 s (Fig. 12(a)) for both cases. By 3 s, some of the aerosol particles activate into cloud droplets and the f_r of aerosol particles broaden for Case L (Fig. 12(b)). Between 3 s to 15 s, particles grow further, and the mean radii of aerosol particles and cloud droplets increase (Fig. 12(c)). The increase of particle radius and broadening of f_r are also evident for Case L-MGF (Figs. 12(d)-(e)). The f_r of cloud droplets and aerosol particles are skewed to the left for Case L at 3 s and 15 s. Thus, most cloud droplets and aerosol particles are larger than their mean values. Interestingly, the f_r of cloud droplets is skewed to the right at 15 s for Case L-MGF (Fig. 12(e)). The maximum radius is much larger for this case ($13.5 \mu m$) than that for Case L ($5.1 \mu m$). Therefore, some cloud droplets experience extreme growth in our model

when mean gradient forcing is applied. The extreme growth has been referred to as super-adiabatic growth⁴³ since the f_r of cloud droplets has broadened more than it would in an adiabatic environment. At the same time, some cloud droplets experience reduced growth for Case L-MGF as the mean and minimum radii of cloud droplets are smaller than those for Case L at 15 s. We also see that the f_r of cloud droplets and aerosol particles overlap (Figs. 12(b)-(e)). For a lognormal dry aerosol size distribution, the corresponding r_c (Eq. (18)) forms a distribution (Figs. 12(b)-(e)) and the activation barrier (r_c) varies from particle to particle. Thus, a smaller particle can be classified as a cloud droplet while a larger particle is still an aerosol, causing an overlap of the respective PDFs. The aerosols and cloud droplets coexist in natural clouds⁴⁴ and our model can distinguish between them under varying conditions.



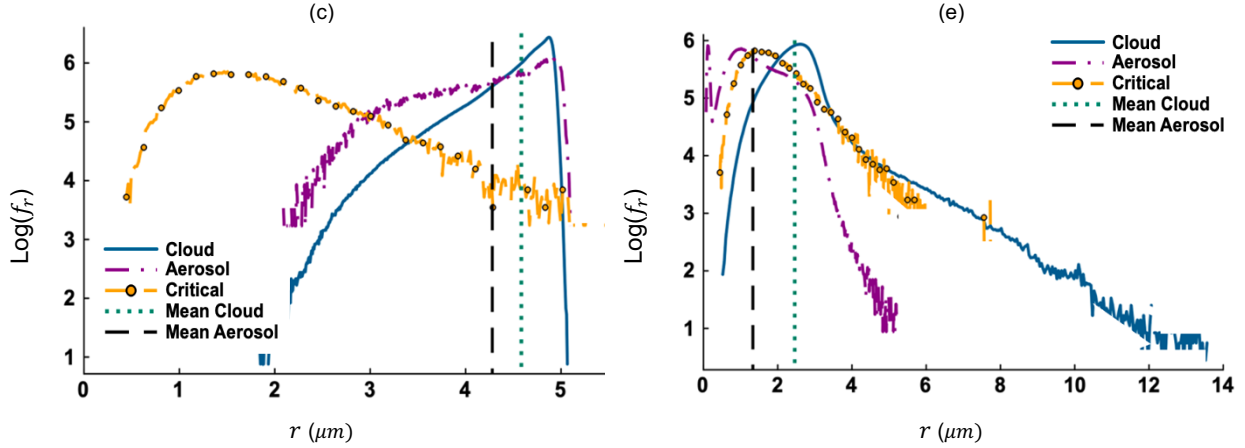


FIG. 12 (a) Log PDF of dry aerosol radii at 0 s. Log PDFs of particle size at (b) 3 s and (c) 15 s for Case L in Study I. Log PDFs of particle size at (d) 3 s and (e) 15 s for Case L-MGF in Study I. The curve with ‘circle’ marker shows the critical radius (r_c) distribution. Scalar forcing is absent in Case L. Case L-MGF applies mean gradient forcing and calculates the β adaptively.

2. Deactivation of cloud droplets (Study II)

The fraction of cloud droplets that deactivate into aerosol particles with time is shown in Fig. 13(a). We see that the deactivation fraction is maximum for the unforced case (L). Figure 13(b) shows the time evolution of mean radius during evaporation. The mean radius decreases most for Case L and least for Case L-MGF. Scalar forcing slows down phase change and evaporative decay in Study II. The profiles in Fig. 13(b) and Fig. 13(a) are apparently in opposite order. As droplet radii decrease by evaporation, the deactivation fraction goes up. The smallest possible radius is the dry radius (r_d) at each particle location. Hence, the mean radius is around $0.1 \mu\text{m}$ and not exactly zero for Case L at long time (Fig. 13(b)). We also see that the mean radius increases for Cases L-MG and L-F after reaching a minimum at approximately 10 s and 15 s, respectively. This coincides with the time equilibrium is reached (Fig. 10(c)). Since $\langle S_e \rangle \sim 0$ at equilibrium, the S'_e is relatively significant. Scalar forcing increases both positive and negative fluctuations in the S_e . But negative fluctuations cannot reduce the particle radius (r) further if it already equals to r_d while positive fluctuations can always make r bigger. Thus, the increase of mean radius for Cases L-MG and L-F corresponds to positive S'_e . With linear forcing (Case L-LF), the increase of r due to positive S'_e balances the decrease of r due to negative S'_e and the mean radius does not change after 15 s (Fig. 13(b)).

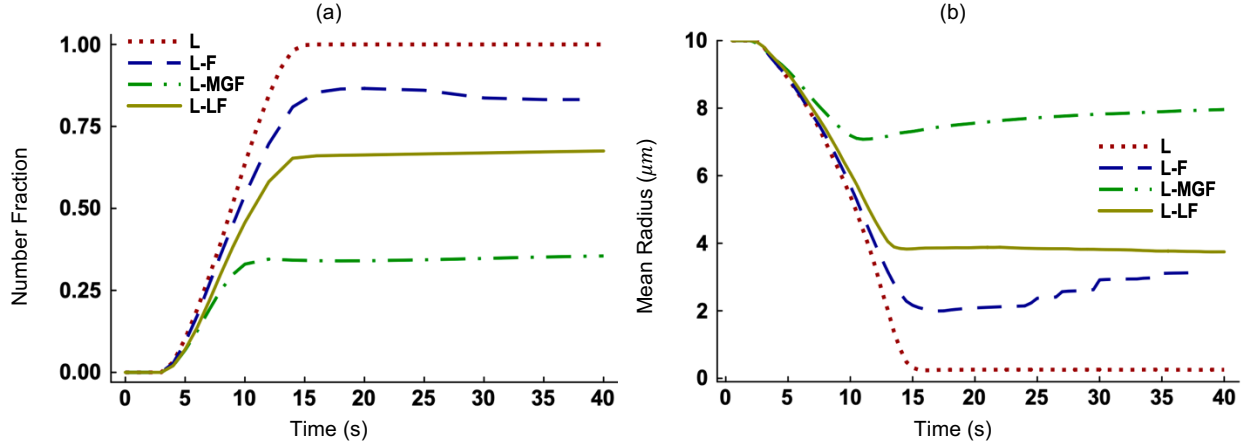


FIG. 13 Time evolution of the (a) number of deactivated cloud droplets, and (b) mean radius of all particles, in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is ‘low’ for all cases.

The PDFs of particle radii (f_r) are shown in Figs. 14(a)-(e) at three selected times in Study II. As dry subsaturated air mixes with the cloud volume, the mean radii of initially monodisperse (10 μm) cloud droplets (Fig. 14(a)) decrease between 0 s to 5 s for Cases L (Figs. 14(b)) and L-LF (Fig. 14(d)). The f_r of cloud droplets broadens and is skewed to the left for both cases at 5 s. On the other hand, the f_r of aerosol particles is skewed to the right. Between 5 s to 15 s, strong evaporation reduces the mean and maximum cloud droplet radii for Case L and the f_r of cloud droplets narrows (Fig. 14(c)). Although the mean droplet radius decreases, the maximum droplet radius does not decrease for Case L-LF at 15 s (Fig. 14(e)) compared to that at 5 s (Fig. 14(d)) and thus narrowing is not observed. It appears that some droplets do not evaporate at all and have radii close to the initial 10 μm while some face significant evaporation and reduction in radius. Hence, cloud droplet evaporation is inhomogeneous⁴⁵ when linear forcing is applied. We see that the f_r of aerosol particles is bimodal for both cases during deactivation, with one peak for smaller aerosol particles and another peak for larger particles (Figs. 14(b)-(e)). But the probability of aerosol particles being small is higher. Note that bimodal size distribution of aerosol particles⁴⁶ has been observed in atmospheric clouds.

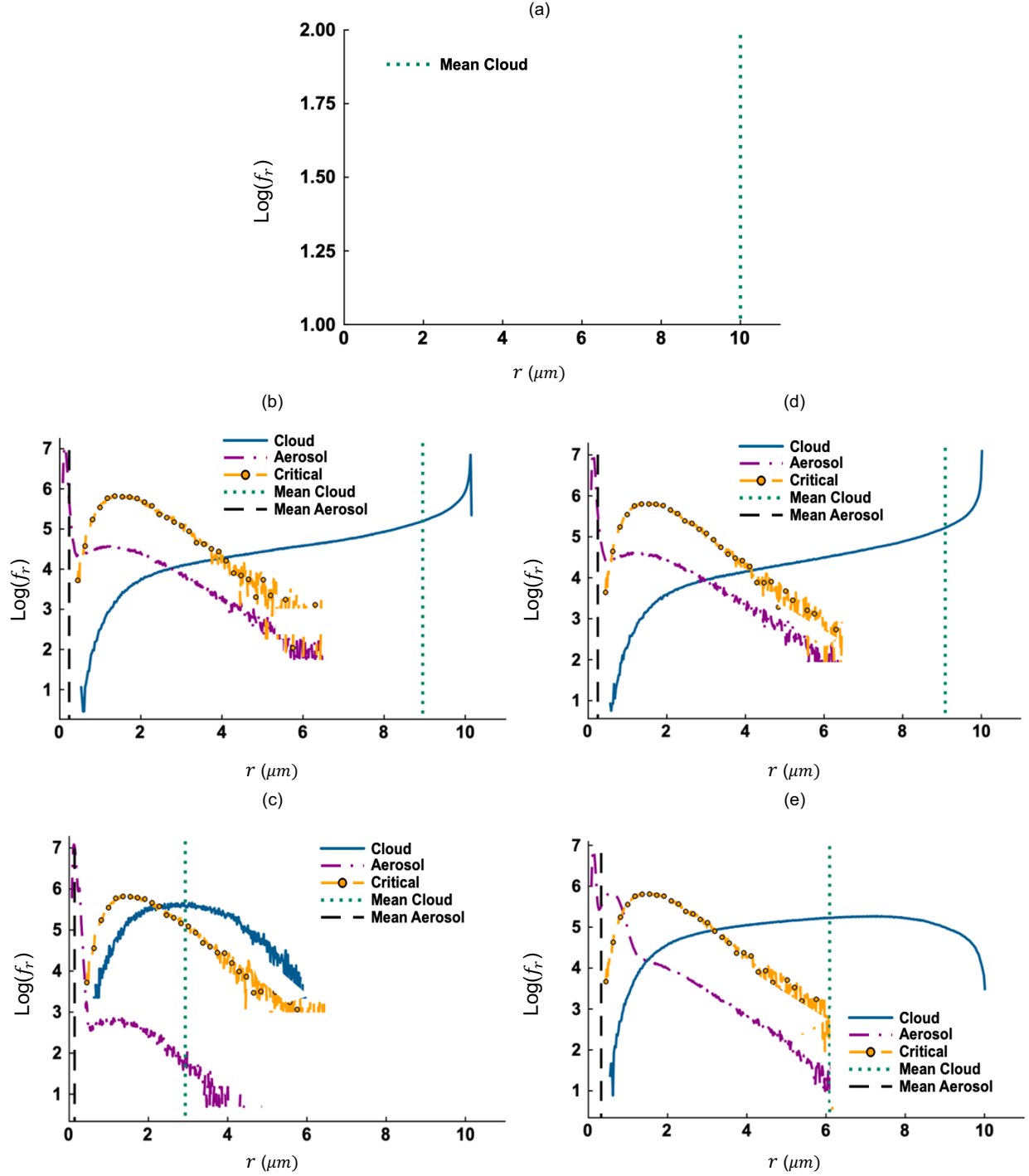


FIG. 14 (a) Log PDF of cloud droplet radii at 0 s. Log PDFs of particle size at (b) 5 s and (c) 15 s for Case L in Study II. Log PDFs of particle size at (d) 5 s and (e) 15 s for Case L-LF in Study II. The curve with ‘circle’ marker shows the critical radius (r_c) distribution. Scalar forcing is absent in Case L. Case L-LF applies linear scalar forcing and calculates the α adaptively.

C. Responses to variation of flow turbulence intensity

We increase the intensity of flow turbulence and simulate Cases H and H-F, where ‘H’ denotes high turbulence level and ‘F’ denotes scalar forcing in the Fourier space. Figure 15(a) compares the variances of vapor mixing ratio σ_{qv}^2 for Cases L, H, L-F, and H-F. The σ_{qv}^2 decays faster for Case H compared to Case L. In absence of scalar forcing, higher velocity turbulence homogenizes the scalar fields⁶ and reduces fluctuations in the temperature, vapor mixing ratio and supersaturation. When scalar forcing is applied (Cases L-F and H-F), it counteracts turbulent homogenization and the σ_{qv}^2 becomes statistically stationary (Fig. 15(a)). The spectra of vapor mixing ratio are plotted in Figure 15(b). They differ at the large and intermediate scales but converge at the small scales for Cases L and H. For the forced cases (L-F and H-F), vapor fluctuations increase significantly at the intermediate and small scales. So, scalar forcing increases small-scale fluctuations while flow turbulence reduces large-scale fluctuations in the thermodynamic fields. The PDFs of fluctuating z-velocity ($f_{w'}$) and fluctuating environmental supersaturation ($f_{s'_e}$) are shown in Figs. 16(a)-(b), respectively. The $f_{w'}$ for Cases H and H-F are broader than those for Cases L and L-F (Fig. 16(a)). Broadening of the z-velocity PDF is caused by a higher flow turbulence. The $f_{w'}$ are identical for Cases L and L-F, and for Cases H and H-F because the scalar fields are passive and do not influence the velocity field in our model.⁶ The PDFs of x- and y-velocities resemble those of z-velocity as turbulence is isotropic. Since fluctuations in the T , q_v , and S_e decrease without forcing, the $f_{s'_e}$ are narrower for Cases L and H compared to the forced cases at 25 s (Fig. 16(b)). The kurtosis of $f_{w'}$ are 2.31, 2.27, 2.40, and 2.42 while that of $f_{s'_e}$ are 3.08, 4.45, 2.40, and 6.21 for Cases L, L-F, H, and H-F, respectively. So, the $f_{s'_e}$ is less Gaussian for Case H (2.40) compared to Case L (3.08) and scalar forcing causes the $f_{s'_e}$ to deviate from a Gaussian profile (3.0). However, neither momentum nor scalar forcing has significant influence on the kurtosis of $f_{w'}$.

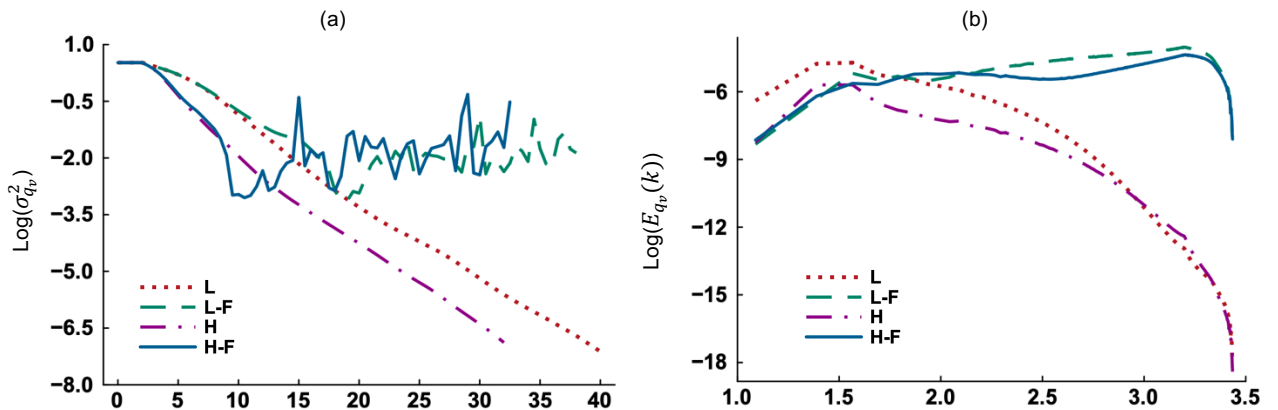


FIG. 15 (a) Time evolution of the variance of fluctuating vapor mixing ratio in Study II and (b) Spectra of fluctuating vapor mixing ratio field at 25 s in Study II. Scalar forcing is absent in Cases L and H. Cases L-F and H-F apply scalar forcing in the Fourier space with ‘low’ and ‘high’ velocity turbulence levels, respectively.

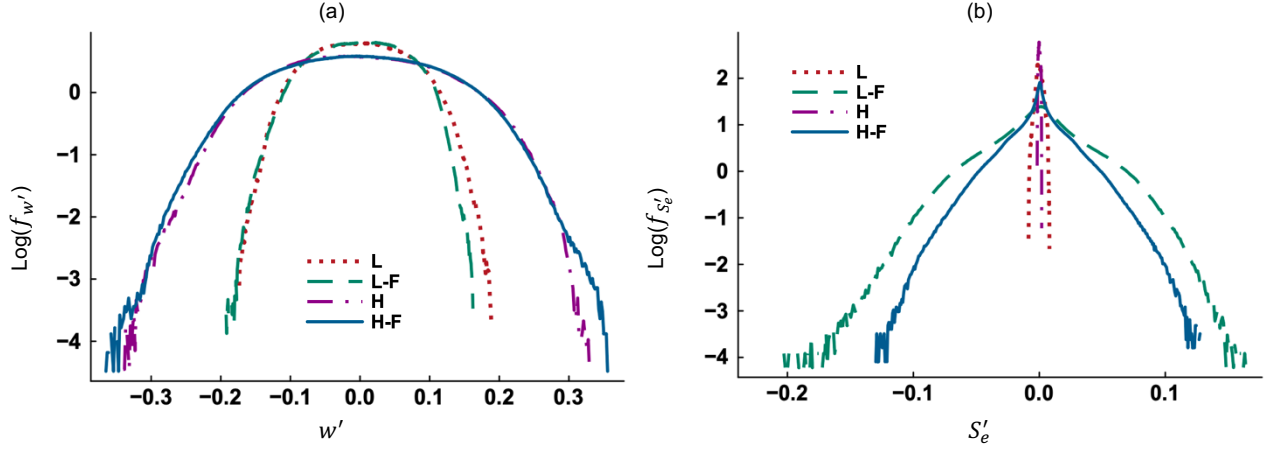


FIG. 16 Log PDFs of the (a) fluctuating z-velocity (w') and (b) fluctuating environmental supersaturation (S'_e) at 25 s in Study II.

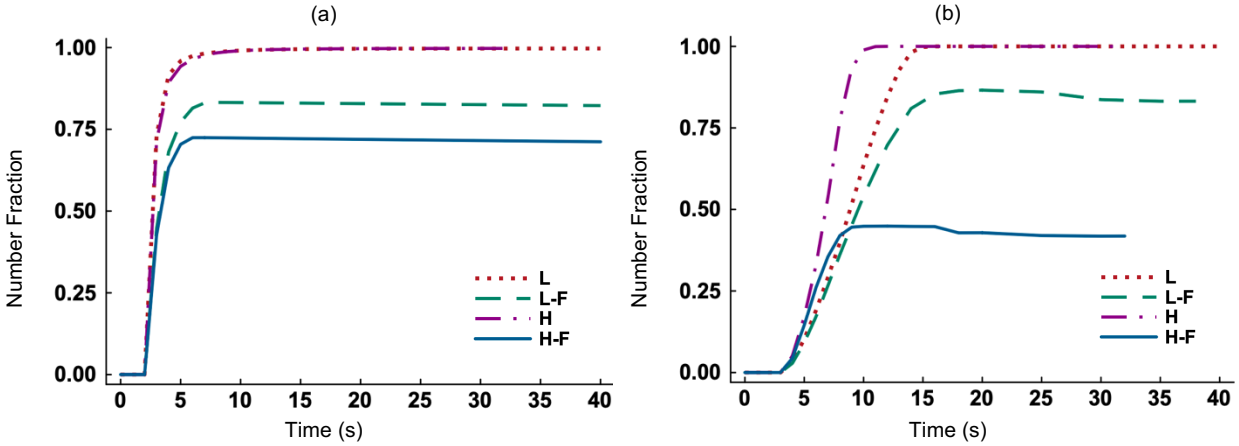


FIG. 17 Time evolution of the (a) number fraction of activated aerosol particles in Study I and (b) number fraction of deactivated cloud droplets in Study II. Scalar forcing is absent in Cases L and H. Cases L-F and H-F apply scalar forcing in the Fourier space with ‘low’ and ‘high’ velocity turbulence levels, respectively.

The magnitudes of momentum and scalar forcings affect the microphysics of aerosols and cloud droplets in our model. Figure 17(a) shows the number fractions of activated aerosol particles for the four cases. The activation fraction is slightly smaller for Case H compared to Case L between 3 s to 7 s. Higher turbulence leads to stronger mixing and causes most particles to grow evenly instead of some particles growing fast and crossing the activation barrier. Thus, activation is delayed if the turbulence level is high and scalar forcing is absent (Case H). But all aerosol particles activate into cloud droplets eventually for the unforced cases. We also notice that the activation fraction is lower for Cases H-F and L-F compared to Cases H and L (Fig. 17(a)). As scalar forcing leads the system towards equilibrium

faster, it suppresses phase change (condensation and activation) in Study I. The deactivation profiles in Fig. 17(b) show that deactivation is faster for Case H compared to Case L. Because of high turbulence and homogenization, more cloud droplets experience partial evaporation rather than some of them evaporating fully. This increases the number of cloud droplets that shrink below the critical radii and deactivate (Case H). Again, scalar forcing slows down phase change (evaporation) and reduces the deactivation fraction in Study II. Both activation and deactivation fractions are smaller for Case H-F compared to Case L-F (Figs. 17(a)-(b)). Scalar forcing suppresses activation and deactivation further if the flow turbulence level is higher.

V. CONCLUDING REMARKS

We have employed a three-dimensional direct numerical simulation (DNS) model, with external forcing in the momentum and scalar fields. There is one-way coupling between turbulent velocity and scalar fields, with the former influencing the latter but not the opposite. There is also two-way coupling between scalar fields and particles. A realistic initial configuration has been used to investigate the effects of scalar forcing on condensation (activation) and evaporation (deactivation). The key findings from our simulations are itemized below.

- The mean and fluctuating scalar fields evolve from nonzero initial conditions. Forcing of the scalar fields leads scalar fluctuations to statistical stationarity (Figs. 3 and 7) and drives mean scalar fields to equilibrium faster (Figs. 5 and 10). Unlike the mean velocity field, the mean scalar fields are not zero in our model. So, forcing affects the evolution of mean scalar fields.
- While dissipation rates in the initial scalar fields are known, the mean scalar gradient β and the scalar variance α are not known a priori. Hence, it is easier to control the magnitude of forcing using the Fourier method than the other two methods. The magnitude of forcing is higher with mean gradient forcing compared to Fourier and linear forcings in our model. The profiles of activation (Fig. 11(a)) and deactivation (Fig. 13(a)) fractions are dissimilar corresponding to three types of scalar forcing.
- The scalar fields are coupled with particles and have reduced diffusivity and higher Schmidt number. Also, their spectral profiles include a viscous-convective regime (Figs. 4(a)-(b)) that falls between the inertial-convective and dissipation ranges. The spectra of unforced and forced scalar fields differ at the intermediate and small scales because forcing makes the viscous-convective regime dominant.

- No phase changes (condensation and evaporation) occur, and the mean vapor mixing ratio remains unchanged with time (Fig. 6(a)) in absence of particles. Scalar forcing itself does not cause deviation from homogeneity and isotropy in the scalar fields should particles be absent (Fig. 6(b)).
- The PDFs of fluctuating scalar fields have high kurtosis and are not Gaussian if forcing is applied (Fig. 9). Mean gradient forcing produces extreme fluctuations in our model and causes super-adiabatic droplet growth (Fig. 12(e)) which has been observed in stratocumulus clouds.⁴³
- Scalar forcing reduces the time available for phase change by leading the system to equilibrium faster. Hence, condensational growth is suppressed (Fig. 11(b)) and fewer aerosol particles activate into cloud droplets (Fig. 11(a)). Similarly, evaporation is suppressed (Fig. 13(b)) and fewer cloud droplets deactivate into aerosol particles (Fig. 13(a)).
- Higher velocity turbulence homogenizes the scalar fields and reduces fluctuations in absence of scalar forcing. On the other hand, scalar forcing counteracts turbulent homogenization (Fig. 15(a)) in our model. Scalar forcing affects thermodynamic fluctuations at the small scales while momentum forcing does so at the large scales (Fig. 15(b)).
- We report in Figs. 12 and 14 that the radius PDFs of cloud are broader at long time compared to early time for both forced and unforced cases. This is because small-scale (scalar forcing) as well as large-scale (momentum forcing) thermodynamic fluctuations contribute to the broadening of radius PDFs. Manton also suggested that both small-scale and large-scale fluctuations in supersaturation broaden the cloud droplet size distributions.⁴⁷
- The magnitude of thermodynamic fluctuations varies in natural clouds.⁷ We show that a DNS model with scalar forcing can maintain thermodynamic fluctuations at desired levels and produce microphysical results that agree with physical observations.

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DATA AVAILABILITY

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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